Kirchhoff calc report

IEC-based, illustrative values, not for construction. Every number comes from the deterministic engine; text was checked against the digit allowlist.
Demo workspace - all companies, people and data are fictional.

Summary

Docs prepared the calc report and BOQ for r2: 150 checks, 0 failing, 2 fixes applied.
Revision r2 · hash cc379d7570ef6aaccb05f6913ff99279675c64fe0e308f8710b301c30fb2e9da · fail 0, n/a 1, pass 149, warn 0

Requirements (tender citations; verification shown per row)

Id	Clause	Page	Value	Unit	Verified
REQ-max_lead_time_weeks	§2.3	2	26	weeks	yes
REQ-rack_count	§3.1	3	48	racks	yes
REQ-rack_kw	§3.1	3	132	kW	yes
REQ-heat_to_liquid_pct	§3.1	3	80	%	yes
REQ-heat_to_air_pct	§3.1	3	20	%	yes
REQ-auxiliary_load_mw	§3.2	3	0.3	MW	yes
REQ-supply_voltage_kv	§3.3	3	33	kV	yes
REQ-supply_frequency_hz	§3.3	3			no
REQ-source_fault_level_mva	§3.3	3	1,000	MVA	yes
REQ-source_rx_ratio	§3.3	3	0.1		yes
REQ-distribution_voltage_v	§3.3	3	415	V	yes
REQ-design_power_factor	§3.3	3	0.95		yes
REQ-critical_power_redundancy	§4.2	4	2N		yes
REQ-transformer_loading_normal_pct	§4.3	4	80	%	yes
REQ-transformer_loading_failure_pct	§4.3	4	100	%	yes
REQ-ups_autonomy_min	§4.5	4	5	minutes	yes
REQ-dv_max_pct	§4.6	4	2	%	yes
REQ-low_level_pipework_elevation_m	§5.1	5	2.6	m	yes
REQ-busway_elevation_m	§5.1	5	3.2	m	yes
REQ-mechanical_mains_elevation_m	§5.1	5	3.8	m	yes
REQ-coolant_supply_temp_c	§6.1	6	30	°C	yes
REQ-coolant_temp_rise_k	§6.1	6	10	K	yes
REQ-cdu_redundancy	§6.2	6	N+1		yes
REQ-pipework_column_clearance_mm	§6.3.4	6	300	mm	yes
REQ-max_coolant_velocity_m_s	§6.4	6	2.5	m/s	yes
REQ-alternative_power_redundancy	App. B	7	Block or distributed redundancy		yes

Topology options

Option	Installed MVA	Normal %	Failover %	Cost USD	Losses kW	Max lead (wk)	SPOF
2N	18.9	38.07	76.15	5,795,160	283.15	14	0
DR	12.6	57.11	76.15	4,517,600	287.6	14	0

Option	Installed MVA	Normal %	Failover %	Cost USD	Losses kW	Max lead (wk)	SPOF
BLOCK	12.6	76.15	76.15	4,672,160	347.42	16	0

Checker findings

Rule	Element	Status	Value	Limit	Unit	Clauses
R01	T-A1	pass	76.15	100	%	§4.3
R01	T-A2	pass	76.15	100	%	§4.3
R01	T-A3	pass	76.15	100	%	§4.3
R01	T-B1	pass	76.15	100	%	§4.3
R01	T-B2	pass	76.15	100	%	§4.3
R01	T-B3	pass	76.15	100	%	§4.3
R02	A-01	pass	1,546	1,600	A	
R02	A-02	pass	1,546	1,600	A	
R02	A-03	pass	1,546	1,600	A	
R02	A-04	pass	1,546	1,600	A	
R02	A-05	pass	1,546	1,600	A	
R02	A-06	pass	1,546	1,600	A	
R02	B-01	pass	1,546	1,600	A	
R02	B-02	pass	1,546	1,600	A	
R02	B-03	pass	1,546	1,600	A	
R02	B-04	pass	1,546	1,600	A	
R02	B-05	pass	1,546	1,600	A	
R02	B-06	pass	1,546	1,600	A	
R02	FA-01	pass	1,546	2,500	A	
R02	FA-02	pass	1,546	1,600	A	
R02	FA-03	pass	1,546	1,600	A	
R02	FA-04	pass	1,546	1,600	A	
R02	FA-05	pass	1,546	1,600	A	
R02	FA-06	pass	1,546	1,600	A	
R02	FB-01	pass	1,546	1,600	A	
R02	FB-02	pass	1,546	1,600	A	
R02	FB-03	pass	1,546	1,600	A	
R02	FB-04	pass	1,546	1,600	A	
R02	FB-05	pass	1,546	1,600	A	
R02	FB-06	pass	1,546	1,600	A	
R03	FA-01	pass	1,600	1,600	A	
R03	FA-02	pass	1,600	1,600	A	
R03	FA-03	pass	1,600	1,600	A	
R03	FA-04	pass	1,600	1,600	A	
R03	FA-05	pass	1,600	1,600	A	
R03	FA-06	pass	1,600	1,600	A	
R03	FB-01	pass	1,600	1,600	A	
R03	FB-02	pass	1,600	1,600	A	
R03	FB-03	pass	1,600	1,600	A	
R03	FB-04	pass	1,600	1,600	A	
R03	FB-05	pass	1,600	1,600	A	
R03	FB-06	pass	1,600	1,600	A	
R04	A1	pass	65.42	100	kA	§4.4

Rule	Element	Status	Value	Limit	Unit	Clauses
R04	A2	pass	65.42	100	kA	§4.4
R04	A3	pass	65.42	100	kA	§4.4
R04	B1	pass	65.42	100	kA	§4.4
R04	B2	pass	65.42	100	kA	§4.4
R04	B3	pass	65.42	100	kA	§4.4
R04	CB-A1	pass	65.42	100	kA	§4.4
R04	CB-A2	pass	65.42	100	kA	§4.4
R04	CB-A3	pass	65.42	100	kA	§4.4
R04	CB-B1	pass	65.42	100	kA	§4.4
R04	CB-B2	pass	65.42	100	kA	§4.4
R04	CB-B3	pass	65.42	100	kA	§4.4
R04b	A-01	pass	52.37	85	kA	§4.4
R04b	A-02	pass	43.57	85	kA	§4.4
R04b	A-03	pass	45.33	85	kA	§4.4
R04b	A-04	pass	47.2	85	kA	§4.4
R04b	A-05	pass	49.19	85	kA	§4.4
R04b	A-06	pass	51.31	85	kA	§4.4
R04b	B-01	pass	47.66	85	kA	§4.4
R04b	B-02	pass	52.62	85	kA	§4.4
R04b	B-03	pass	54.66	85	kA	§4.4
R04b	B-04	pass	53.97	85	kA	§4.4
R04b	B-05	pass	51.96	85	kA	§4.4
R04b	B-06	pass	50.05	85	kA	§4.4
R04b	FA-01	n/a	n/a	n/a		§4.4
R04b	FA-02	pass	65.42	85	kA	§4.4
R04b	FA-03	pass	65.42	85	kA	§4.4
R04b	FA-04	pass	65.42	85	kA	§4.4
R04b	FA-05	pass	65.42	85	kA	§4.4
R04b	FA-06	pass	65.42	85	kA	§4.4
R04b	FB-01	pass	65.42	85	kA	§4.4
R04b	FB-02	pass	65.42	85	kA	§4.4
R04b	FB-03	pass	65.42	85	kA	§4.4
R04b	FB-04	pass	65.42	85	kA	§4.4
R04b	FB-05	pass	65.42	85	kA	§4.4
R04b	FB-06	pass	65.42	85	kA	§4.4
R05	FA-01	pass	0.84	2	%	§4.6
R05	FA-02	pass	1.73	2	%	§4.6
R05	FA-03	pass	1.56	2	%	§4.6
R05	FA-04	pass	1.39	2	%	§4.6
R05	FA-05	pass	1.21	2	%	§4.6
R05	FA-06	pass	1.04	2	%	§4.6
R05	FB-01	pass	1.34	2	%	§4.6
R05	FB-02	pass	0.94	2	%	§4.6
R05	FB-03	pass	0.78	2	%	§4.6
R05	FB-04	pass	0.83	2	%	§4.6
R05	FB-05	pass	0.99	2	%	§4.6
R05	FB-06	pass	1.14	2	%	§4.6
R06	HALL	pass	0	0		§4.2

Rule	Element	Status	Value	Limit	Unit	Clauses
R07	UPS-A1	pass	2,157	2,500	kW	§4.5
R07	UPS-A2	pass	2,157	2,500	kW	§4.5
R07	UPS-A3	pass	2,157	2,500	kW	§4.5
R07	UPS-B1	pass	2,157	2,500	kW	§4.5
R07	UPS-B2	pass	2,157	2,500	kW	§4.5
R07	UPS-B3	pass	2,157	2,500	kW	§4.5
R08	Z1	pass	1,690	2,400	kW	§6.2
R08	Z2	pass	1,690	2,400	kW	§6.2
R08	Z3	pass	1,690	2,400	kW	§6.2
R09	M-R1	pass	157.26	157.26	L/min	§6.1
R09	M-R2	pass	157.26	157.26	L/min	§6.1
R09	M-R3	pass	157.26	157.26	L/min	§6.1
R09	M-R4	pass	157.26	157.26	L/min	§6.1
R09	M-R5	pass	157.26	157.26	L/min	§6.1
R09	M-R6	pass	157.26	157.26	L/min	§6.1
R10	BR-R1	pass	1.13	2.5	m/s	§6.4
R10	BR-R2	pass	1.13	2.5	m/s	§6.4
R10	BR-R3	pass	1.13	2.5	m/s	§6.4
R10	BR-R4	pass	1.13	2.5	m/s	§6.4
R10	BR-R5	pass	1.13	2.5	m/s	§6.4
R10	BR-R6	pass	1.13	2.5	m/s	§6.4
R10	H-Z1	pass	1.26	2.5	m/s	§6.4
R10	H-Z2	pass	1.26	2.5	m/s	§6.4
R10	H-Z3	pass	1.26	2.5	m/s	§6.4
R11	BR-R1	pass	5.8	0.3	m	§6.3.4
R11	BR-R2	pass	3.33	0.3	m	§6.3.4
R11	BR-R3	pass	3.33	0.3	m	§6.3.4
R11	BR-R4	pass	1.8	0.3	m	§6.3.4
R11	BR-R5	pass	4.72	0.3	m	§6.3.4
R11	BR-R6	pass	3.33	0.3	m	§6.3.4
R11	H-Z1	pass	3.8	0.3	m	§6.3.4
R11	H-Z2	pass	1.8	0.3	m	§6.3.4
R11	H-Z3	pass	4.8	0.3	m	§6.3.4
R12	BOQ	pass	0	0	lines	
R13	CDU-1	pass	2	2	systems	§6.5
R13	CDU-2	pass	2	2	systems	§6.5
R13	CDU-3	pass	2	2	systems	§6.5
R13	CDU-4	pass	2	2	systems	§6.5
R13	CDU-5	pass	2	2	systems	§6.5
R13	CDU-6	pass	2	2	systems	§6.5
R13	CDU-7	pass	2	2	systems	§6.5
R13	CDU-8	pass	2	2	systems	§6.5

Rule	Element	Status	Value	Limit	Unit	Clauses
R13	CDU-9	pass	2	2	systems	§6.5
R14	H-Z1	pass	69.3	150	kPa	§6.2
R14	H-Z2	pass	67.04	150	kPa	§6.2
R14	H-Z3	pass	64.79	150	kPa	§6.2
R15	ACB-1600	pass	0	26	weeks	§2.3
R15	ACB-4000	pass	0	26	weeks	§2.3
R15	ACB-5000	pass	0	26	weeks	§2.3
R15	BW-1600	pass	0	26	weeks	§2.3
R15	CBL-240-8P	pass	0	26	weeks	§2.3
R15	CDU-1200	pass	14	26	weeks	§2.3
R15	CRB-415	pass	0	26	weeks	§2.3
R15	LVB-415	pass	12	26	weeks	§2.3
R15	MVS-33	pass	0	26	weeks	§2.3
R15	PIPE-DN150	pass	0	26	weeks	§2.3
R15	PIPE-DN200	pass	0	26	weeks	§2.3
R15	TX-3150	pass	0	26	weeks	§2.3
R15	UPS-500	pass	0	26	weeks	§2.3

Fixes

R05:FA-01: Replace FA-01 with a parallel cable feeder (Cable feeder 8 x 1C 240 mm2 Cu per phase) (power). Candidate FA-01:cable-CBL-240-8P is selected because it provides the largest voltage-drop margin per dollar for FA-01 under rule R05. Formula: dU% = 100·sqrt3·I·sum L·(R·cosphi + X·sinphi)·k/U; k = 1 (feeder), (n+1)/(2n) (row busway). Rejected: Upsize FA-01 to 2,000 A busway.
R11:H-Z2: Drop H-Z2 to pipe_low where it crosses FB-03 (cooling). Candidate H-Z2:local-drop is selected based on the criterion of fewest changed pipe segments to achieve the least disruption to coordinated routes. Formula: no pipe segment on a tier above the busway tier over an electrical run; clearance >= c_min. Rejected: Reroute H-Z2 clear of FB-03.

Assumptions ledger

[did] 2N: TX-2500 replaced by TX-3150 (R15)
[did] DR: TX-2500 replaced by TX-3150 (R15)
[not_checked] Earth-fault loop impedance (IEC 60364-4-41 §411)
[not_checked] Selectivity between protective devices
[not_checked] Arc flash
[not_checked] Harmonics
[not_checked] Air-side cooling for the air-cooled share of rack heat
[not_checked] MV bus faults
[not_checked] Busway derating above 35 °C ambient (ambient assumed <= 35 °C)
[not_checked] Overload protection (R03) of transformer incomer and UPS breakers: sized to equipment rating, checked by R04 only
[not_checked] Short-circuit withstand of cable feeders
[not_checked] Spoken Gemini Live audio is not digit-filtered; on-screen transcripts are.

Bill of quantities

Material	Description	Qty	Unit price USD	Lead (wk)
ACB-1600	ACB 1,600 A frame, Icu 100 kA	12	9,800.00	0
ACB-4000	ACB 4,000 A frame, Icu 100 kA	12	21,000.00	0
ACB-5000	ACB 5,000 A frame, Icu 100 kA	6	26,500.00	0
BW-1600	Busway 1,600 A, 415 V	536	420.00	0
CBL-240-8P	Cable feeder 8 x 1C 240 mm2 Cu per phase	82	610.00	0
CDU-1200	Coolant distribution unit 1,200 kW	9	145,000.00	14

Material	Description	Qty	Unit price USD	Lead (wk)
CRB-415	Critical (UPS output) board 415 V	6	64,000.00	0
LVB-415	LV main board 415 V	6	85,000.00	12
MVS-33	33 kV switchboard, 2 incomers + bus-coupler	1	240,000.00	0
PIPE-DN150	Steel pipe DN150	68	140.00	0
PIPE-DN200	Steel pipe DN200	56	210.00	0
TX-3150	Cast-resin transformer 3.15 MVA 33/0.415 kV Dyn11	6	118,000.00	0
UPS-500	UPS module 500 kW, 415 V	30	62,000.00	0